

JACKSON MORRIS

CURRICULUM VITAE

Department of Mathematics
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EMPLOYMENT

Postdoctoral Researcher

Vanderbilt University

January 2027-August 2029

Postdoc SLMath

SLMath Semester Program

Motivic Homotopy Theory: Connections and Applications

August-December 2026

EDUCATION

Ph.D. in Mathematics

University of Washington

Advisors: Kyle Ormsby and John Palmieri

Thesis title: Cooperations in motivic homotopy theory

September 2021-June 2026 (Expected)

M.S. in Mathematics

University of Washington

September 2021-June 2023

B.S. in Mathematics, *magna cum laude*

University of Kentucky

September 2017-December 2020

RESEARCH INTERESTS

Motivic homotopy theory, chromatic homotopy theory, hermitian K-theory, algebraic K-theory, synthetic spectra, nilpotence and periodicity, Adams spectral sequence, periodic self-maps, redshift and blueshift

PUBLICATIONS AND PREPRINTS

8. *Diachromatic blueshift*, joint with Kyle Ormsby. In preparation.
7. *Motivic K-theory cooperations over the rationals*, joint with Sarah Petersen and Liz Tatum. In preparation.
7. *Chromatic structures in motivic homotopy theory*. In preparation.
6. *An exotic self-map of periodicity 1*. In preparation.
5. *Immersion of C_2 -projective spaces via $K\mathbb{R}$ -theory*, joint with Manyi Guo, Alex Waugh, and Albert Jinghui Yang. Available at <https://arxiv.org/abs/2604.25260>.
4. *Spectrum level splittings of truncated motivic Brown-Peterson cooperations algebras*, joint with Sarah Petersen and Elizabeth Tatum. Submitted. Available at <https://arxiv.org/abs/2509.19542>.
3. *Rings of cooperations for hermitian K-theory over finite fields*. Submitted. Available at <https://arxiv.org/abs/2509.02786>.
2. *On the ring of cooperations for real hermitian K-theory*. To appear in **Annals of K-theory**. Available at <https://arxiv.org/abs/2506.16672>.
1. *Toric Double Determinantal Varieties*, joint with Alexander Blose, Patricia Klein, and Owen McGrath. **Communications in Algebra** 49 (2020), 7, 3085-3093.

SELECTED PRESENTATIONS & SEMINAR TALKS

Colloquium Talks

“Finding Fixed Points with Invariants”

Reed College Mathematics Colloquium

February 2025

Research Talks

“Higher Witt K-theories”

AMS Sectional: Computational Homotopy Theory (Tempe, AZ)

November 2026

Young Topologists Meeting 2026 (Copenhagen, DK)

July 2026

Cascade Topology Seminar (Portland, OR)

May 2026

“Splittings of truncated motivic Brown–Peterson cooperations algebras”

University of Washington Topology Seminar

February 2026

University of Colorado Boulder Homotopy Theory Seminar

September 2025

“Splittings and cooperations in motivic homotopy theory”

University of Illinois Champaign-Urbana Topology Seminar

January 2026

Joint Math Meetings 2026 (Washington, D.C.)

January 2026

University of Virginia Topology Seminar

November 2025

University of Kentucky Topology Seminar

November 2025

Duke University Geometry and Topology Seminar

October 2025

University of Notre Dame Topology Seminar

September 2025

“On the ring of cooperations for real hermitian K-theory”

University of Michigan Geometry Seminar

September 2025

European Talbot 2025 (Kolding, DK)

July 2025

University of Washington Topology Seminar

June 2025

Graduate Student Geometry & Topology Conference 2025 (Bloomington, IN)

April 2025

Joint Math Meetings 2025 (Seattle, WA)

January 2025

European Autumn School in Topology (Utrecht, NL)

September 2024

Motivic Homotopy, K-theory, and Modular Representations (Los Angeles, CA)

August 2024

“Symplectic Orientations”

GROOT Seminar (Online)

August 2024

Contributed Talks

eCHT Reading Seminars

“An upper bound on the degree of symmetry of an exotic sphere”

March 2026

“Higher Adams differentials and hidden extensions”

November 2025

“Motivic twisted K-theory”

April 2024

Graduate Student Seminars

“Higher real K-theories, redshift, and blueshift”

May 2026

“Filtered objects, spectral sequences, and t -structures”

January 2026

“(A synthetic approach to detecting) v_1 -periodic families”

November 2025

“Algebraic and Milnor K-theory”

October 2025

“Galois descent and the Picard group of K-theory”

May 2025

“The Adams spectral sequence and Hopf algebroids”

February 2025

Workshops

“ $\pi_*\mathrm{THH}(\mathbb{Z}_p^{\mathrm{fil}})/p$ and $\pi_*\mathrm{THH}(\ell_p^{\mathrm{fil}})/(p, v_1)$ ” (Talbot, Nacogdoches TX)

August 2024

“Hochschild Homology” (European Autumn School in Topology, Utrecht, NL)

September 2024

SERVICE AND PROFESSIONAL ACTIVITIES

Refereeing and Reviewing

Algebraic and Geometric Topology

2025–Present

ZBMath

2026–Present

Seminar Organization

DUBTOP Seminar

2025–2026

Topics included: Chromatic homotopy theory, stacks, Goodwillie calculus

Back-To-School seminar

Fall 2024

Student lectures on fun math learned over the summer.

Reading seminars

2023-2024

Topics included: Algebraic K-theory, cohomology operations, motivic homotopy theory, vector bundles and topological K-theory, categorical homotopy theory, model categories

TEACHING EXPERIENCE

Instructor of Record

University of Washington

Math 480 – Homotopy theory

Spring 2026

Math 208 – Linear algebra

Winter 2023, Summer 2023

Graduate Assistant

Electronic Computational Homotopy Theory (eCHT)

Research Seminar

2025-2026

Quadratic Curve Counting

Fall 2024

Park City Math Institute (PCMI)

Summer 2024

Experimental Math Lab

University of Washington

Fall 2021 - Spring 2026

Courses included: Precalculus, Differential calculus, Multivariable calculus, Linear algebra, Introduction to proofs, Topology

Directed Reading Mentor

University of Washington

2022-2025

Topics included: Homotopy theory, Representation theory, Geometric group theory, Homological algebra

Math Olympiad Judge

UW Math Hour

2023 - 2024

Judge for elementary school math competition

HONORS AND AWARDS

Lee and Cozette McFarlan Travel Fellowship

Summer 2025

University of Washington

\$750

For travel to European Talbot in Kolding, DK.

Research Assistantship

Summer 2025

University of Washington

For good research progress.

JC Eaves Scholarship

2020

University of Kentucky

\$2000

For continued excellence in mathematics as a Kentuckian.

Carolyn Bunyan Award

2020

University of Kentucky

\$2000

For mathematics majors interested in continuing education in mathematics.

SELECTED WORKSHOP AND CONFERENCE PARTICIPATION

Extended Research Visits

SLMath Semester Program: Motivic Homotopy Theory (Berkeley, CA)

Fall 2026

PCMI: Motivic Homotopy Theory (Park City, UT)

Summer 2024

Conference Participation

AMS Sectional: Computational Homotopy Theory (Tempe, AZ)

November 2026

Algebraic Structures in Topology (San Juan, PR)

July 2026

Young Topologists Meeting (Copenhagen, DK)	<i>June/July 2026</i>
Cascade Topology Seminar (Portland, OR)	<i>May 2026</i>
(∞, ∞)-Categories Workshop (Chicago, IL)	<i>March 2026</i>
JMM 2026 (Washington, D.C.)	<i>January 2026</i>
Midwest Topology Seminar (Minneapolis, MN)	<i>October 2025</i>
European Talbot (Kolding, DK)	<i>July 2025</i>
WAGS Spring 2025 (Vancouver, CAN)	<i>April 2025</i>
GSTGC 2025 (Bloomington, IN)	<i>April 2025</i>
JMM 2025 (Seattle, WA)	<i>January 2025</i>
SLMath Hot Topics: Life after the Telescope Conjecture (Berkeley, CA)	<i>December 2024</i>
European Autumn School in Topology (Utrecht, NL)	<i>September 2024</i>
Talbot 2024 (Nacogdoches, TX)	<i>August 2024</i>
Motivic Homotopy, K-theory, and Modular Representations (Los Angeles, CA)	<i>August 2024</i>
eCHT Research Workshop on Hopf Rings (Online)	<i>June 2024</i>
JMM 2024 (San Francisco, CA)	<i>January 2024</i>
Scissors Congruence, Algebraic K-theory, & Trace Methods (Bloomington, IN)	<i>July 2023</i>
Cascade Topology Seminar (Vancouver, CAN)	<i>April 2023</i>
PRIMA Congress (Vancouver, CAN)	<i>December 2022</i>
Graduate Students in Geometry, Topology, and Algebra (Philadelphia, PA)	<i>May 2022</i>

REFERENCES

Mark Behrens John and Margaret McAndrews Professor of Mathematics University of Notre Dame <i>Relationship: Research mentor</i>	mbehren1@nd.edu
Kyle Ormsby Mathematics Department Chair Reed College <i>Relationship: PhD advisor and research collaborator</i>	ormsbyk@reed.edu
John Palmieri Professor of Mathematics University of Washington <i>Relationship: PhD advisor; Mathematics Department Chair during first 2 years of graduate study</i>	jpalmier@uw.edu
J.D. Quigley Assistant Professor of Mathematics University of Virginia <i>Relationship: Research mentor; Program supervisor and mentor at PCMI</i>	mbp6pj@virginia.edu
Charles Camacho Assistant Teaching Professor University of Washington <i>Relationship: Teaching supervisor and mentor</i>	camachoc@uw.edu
Natalie Naehrig Assistant Teaching Professor University of Washington <i>Relationship: Teaching supervisor and mentor</i>	naehrn@uw.edu